

Autonomous Shuttle and BRT Integration: An Agent-Based Case Study of Paris’s Southern Suburbs

Laura C. Echeverri^{*1} and Nicolas Coulombel²

¹LVMT, Univ Gustave Eiffel, ENPC, Institut Polytechnique de Paris, Marne-la-Vallée, France

²LVMT, ENPC, Institut Polytechnique de Paris, Univ Gustave Eiffel, Marne-la-Vallée, France

1 INTRODUCTION

Shared autonomous vehicles (SAVs) offer a promising solution for addressing mobility challenges by reducing labor costs, extending service hours, and optimizing public transport (PT) service allocation. Rather than competing with existing transit, SAVs can complement PT systems by taking advantage of their flexibility for low-demand areas or off-peak hours, while preserving PT’s high capacity for dense corridors. This approach could help reallocate underutilized transit resources and enhance overall network efficiency (Carrese et al., 2023). This study examines this potential through the MOB-AUTO² project, which integrates 12-seat autonomous shuttles (ASS) with the *Car Express* 91-03 Bus Rapid Transit (BRT) line in Paris’s southern suburbs.

The BRT line connects Dourdan—a peri-urban area with limited rail access—to Massy – Palaiseau, a major transit hub. While successful in reducing private car use, particularly in Briis-sous-Forges and Dourdan, the line faces capacity constraints during peak hours and limited off-peak service (Coldefy & Gendre, 2020). We evaluate whether ASs operating between Longvilliers and Massy (three out of four BRT stops) could enhance the area’s public transit service by alleviating the load on the BRT line, improving connectivity during periods of low demand, and encouraging new users who previously relied on private cars to switch to public transport. We compare operational performance and perform a cost-benefit analysis before and after integration using agent-based simulation.

The analysis centers on a primary analysis area defined by the residential and workplace locations of *Car Express* users, derived from a synthetic population. To ensure comprehensive coverage of activity centers, we used an extreme simulation scenario (1-minute bus headway). The analysis area is shown in Figure 1.

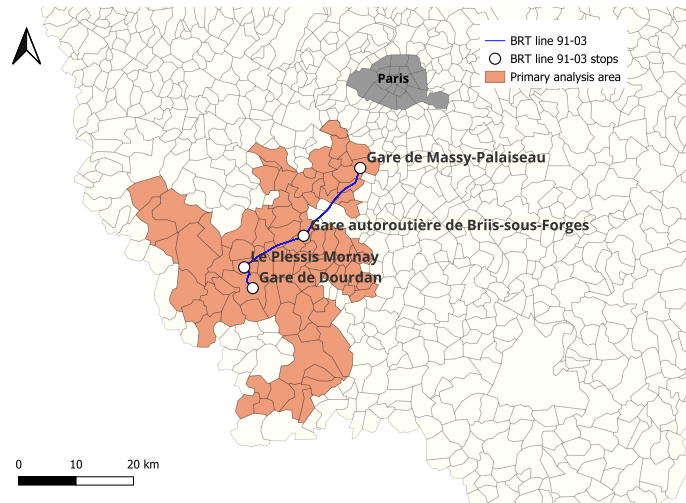


Figure 1: Primary analysis area in Île-de-France region

2 METHODOLOGY

Simulation environment

This study employs the agent-based simulation framework MATSim (Horni et al. (2016)), extended with the Discrete Mode Choice (DMC) module (Hörl & Balac, 2021), to model travel demand via a synthetic population. Each agent is characterized by socio-demographic attributes and a daily activity patterns. The synthetic population (representing 10% of the real population for computational efficiency) and the multimodal transport network were generated using the open-source eqasim pipeline (Hörl & Balac, 2021), which calibrates travel demand and transport supply for the Île-de-France region using public demographic and mobility datasets.

The AS service was implemented in MATSim using two alternative approaches: as a scheduled public transport service integrated into the existing transit network, or as a demand-responsive transport (DRT) service modelled using the DRT module with a stop-based operational scheme (Maciejewski & Nagel, 2013). The choice of approach depends on the scenario: the former is used for PT integration, while the latter is used for competitive evaluations against the BRT line.

Scenario design

To evaluate the integration of a 12-seat AS service between Longvilliers and Massy, we design four scenarios, including a baseline (Business as Usual) and three alternative configurations reflecting different levels of substitution of the existing BRT line 91-03:

- **Business as usual (BAU):** the existing BRT line 91-03 operates without any modification, serving as a reference for performance comparisons.
- **Complementary service (CS):** the AS supplements the BRT line 91-03, doubling the frequency during off-peak hours and alleviating congestion during peak hours. Two AS departures are added between each BRT departure.
- **Partial substitution (PS):** ASs replace low-frequency BRT trips during off-peak periods while maintaining BRT service during peaks to handle high demand. AS departures occur every 5 minutes during regular BRT service hours.
- **Total substitution (TS):** ASs fully replaces the BRT line 91-03, operating as a stop-based Demand-Responsive Transit (DRT) with fixed routes but flexible scheduling.

3 PRELIMINARY RESULTS

We first validated the BAU scenario by comparing simulated modal shares with empirical data from the Enquête Globale Transport (EGT) 2010 (Table 1), revealing strong alignment between the two. Further validation against real-world boarding data (Tuesdays/Thursdays, March–June 2024) showed that the simulated boardings (3,570) closely matched the mean of the observed values (3,202), with a modest overestimation of 11.8%, confirming the model’s reliability for scenario analysis.

	Car	Car passenger	PT	Walk	Bike
Travel survey (EGT 2010)	49%	13%	12%	23%	2%
BAU scenario	44%	13%	9%	31%	2%

Table 1: Comparison of modal share for the primary analysis area

We then compared the results across three scenarios: BAU, CS, and PS. The analysis revealed minimal changes in modal shares across scenarios, indicating that more substantial interventions may be required to achieve significant modal shifts away from private cars. Figure 2 illustrates total boardings on the BRT line and AS service in 30-minute intervals, corresponding to the BRT’s end-to-end travel time. Shaded areas indicate periods where total boardings exceeded the BAU baseline. While the PS scenario shows modest improvements in ridership, it also exhibits user losses during the latter half of peak hours. Overall, these findings suggest that the current configurations in both CS and PS scenarios result in only limited ridership shifts.

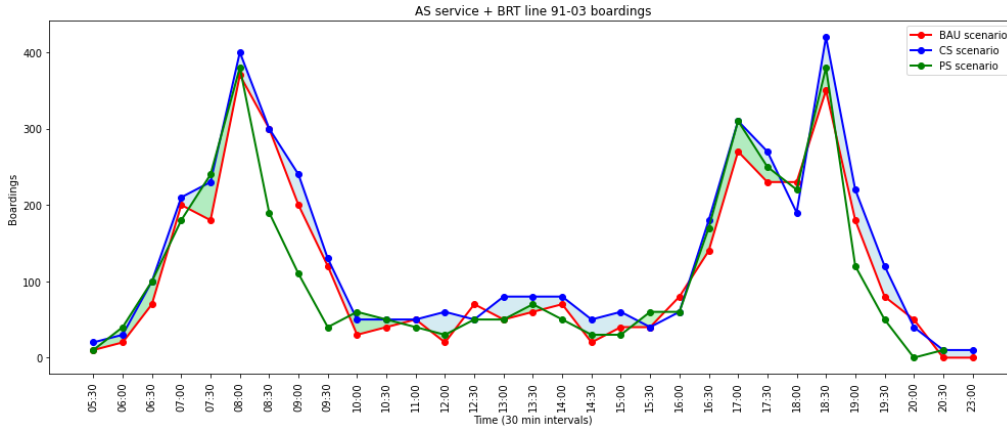


Figure 2: Number of boardings comparison between scenarios

Future work will refine the AS schedules for CS and PS scenarios and analyze the TS scenario, where AS operates as Demand-Responsive Transport (DRT). Additional metrics—including operational performance (e.g., wait times, load factors) and a comprehensive cost-benefit analysis—will determine the optimal AS-BRT integration strategy for this case study.

ACKNOWLEDGEMENTS

This work was carried out as part of the MOB-AUTO² (MOBilite AUTOMatisée sur AUTOroute connectée) project, funded under the France 2030 program.

REFERENCES

- Carrese, F., Sportiello, S., Zhaksylykov, T., Colombaroni, C., Carrese, S., Papaveri, M., & Patella, S. M. (2023). The Integration of Shared Autonomous Vehicles in Public Transportation Services: A Systematic Review. *Sustainability*, 15(17), 13023.
- Coldefy, J., & Gendre, P. (2020, August). *Retours d’expériences cars express* (Tech. Rep.). Comité Scientifique France Mobilités – ATEC ITS France.
- Hörl, S., & Balac, M. (2021). Synthetic population and travel demand for Paris and Île-de-france based on open and publicly available data. *Transportation Research Part C: Emerging Technologies*, 130, 103291.
- Horni, A., Nagel, K., & Axhausen, K. W. (Eds.). (2016). *The Multi-Agent Transport Simulation MATSim*. London: Ubiquity Press. Retrieved from <https://doi.org/10.5334/baw> doi: 10.5334/baw
- Maciejewski, M., & Nagel, K. (2013). Simulation and Dynamic Optimization of Taxi Services in matsim. *Transportation Science*, 10-13, 34.